

Fig. 5: *An example of a plot of sediment load against overland flow depth, showing the least-squares rating equation constrained to pass through the origin.*

Experiment	Minute	Sediment load S1 ($g\ m^{-1}\ min^{-1}$)	Sediment load S2 ($g\ m^{-1}\ min^{-1}$)	Average soil loss S1 ($g\ m^{-2}\ min^{-1}$)	Average soil loss S2 ($g\ m^{-2}\ min^{-1}$)
1	4	5.13	15.87	0.41	0.76
	9	9.83	14.67	0.79	0.70
2	4	11.27	11.40	0.90	0.54
	9	14.50	28.93	1.16	1.38
	14	15.57	31.73	1.25	1.51
	19*	9.70	24.00	0.78	1.14
3	4	17.97	23.87	1.44	1.14
	9	16.27	22.97	1.30	1.09
4	4	9.23	18.47	0.74	0.88
	9	7.00	15.53	0.56	0.74
	14	9.10	17.23	0.73	0.82
	19*	6.70	19.37	0.54	0.92
	24*	8.00	13.67	0.64	0.65
	29*	8.30	14.87	0.66	0.71

* Equilibrium runoff conditions

Tab. 1: *Sediment loads and average soil losses.*

tions usually overestimated the sediment loads in the shallowest flows. Inasmuch as the depth distributions were negative exponential, overestimation of the loads in these flows often resulted in the gross overestimation of the section sediment load. The most consistent section sediment loads were obtained when sediment load was plotted against flow depth on arithmetic graph paper, and the rating equation was forced to pass through the origin (fig. 5). The section sediment loads (per unit width) T obtained by combining these rating equations with their associated depth distributions are listed in tab. 1.